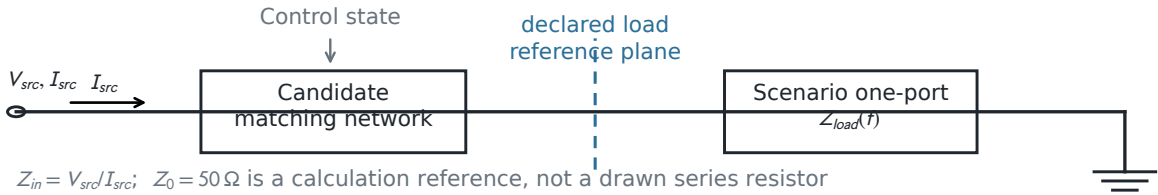


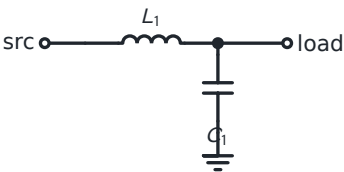
RF analysis boundary and matching topologies

Connections used by the public circuit builders; the chamber enters only as a declared electrical one-port.

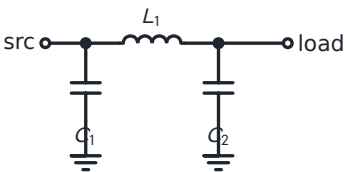
(a) Analysis boundary



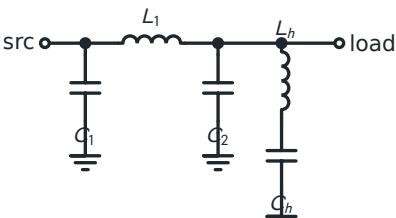
(b) L-match



(c) Pi-match



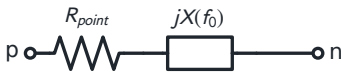
(d) Pi-match + harmonic branch



Electrical load models used by the core benchmarks

Each load is a named-reference-plane one-port; geometry, plasma state and species-power partition remain outside scope.

(a) Impedance point



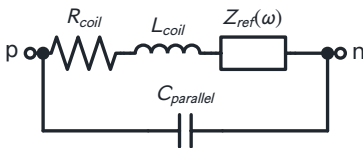
$X > 0$: $L = X/\omega_0$ $X < 0$: $C = -1/(\omega_0 X)$

(b) Effective CCP one-port



$Z = R_{eff} + j\omega L_{eff} + 1/(j\omega C_{sheath, eq})$

(c) Reduced ICP coil one-port

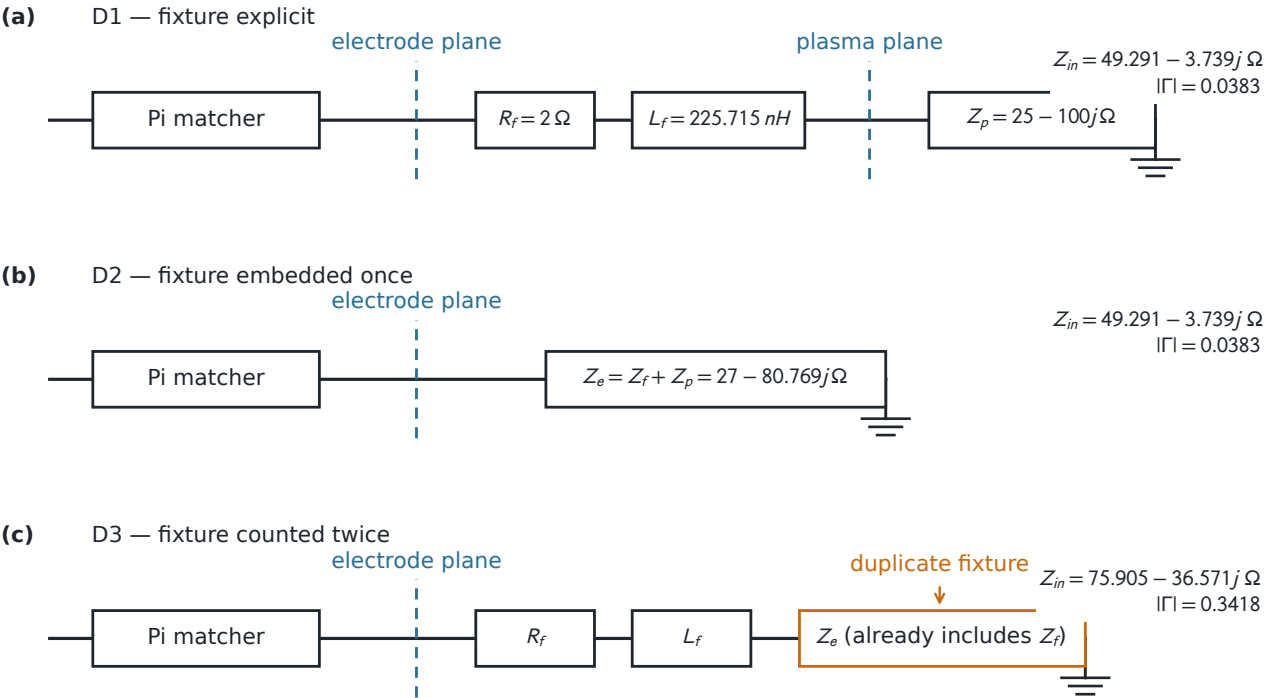


$Z_{ref} = \omega^2 L_{reflected} / (\gamma + j\omega)$

Scope: A4 checks the series CCP terminal form; A5 checks the identifiable ICP reduction.
ICP transformer lineage: Lee et al. (2020), DOI 10.1063/1.5133862.

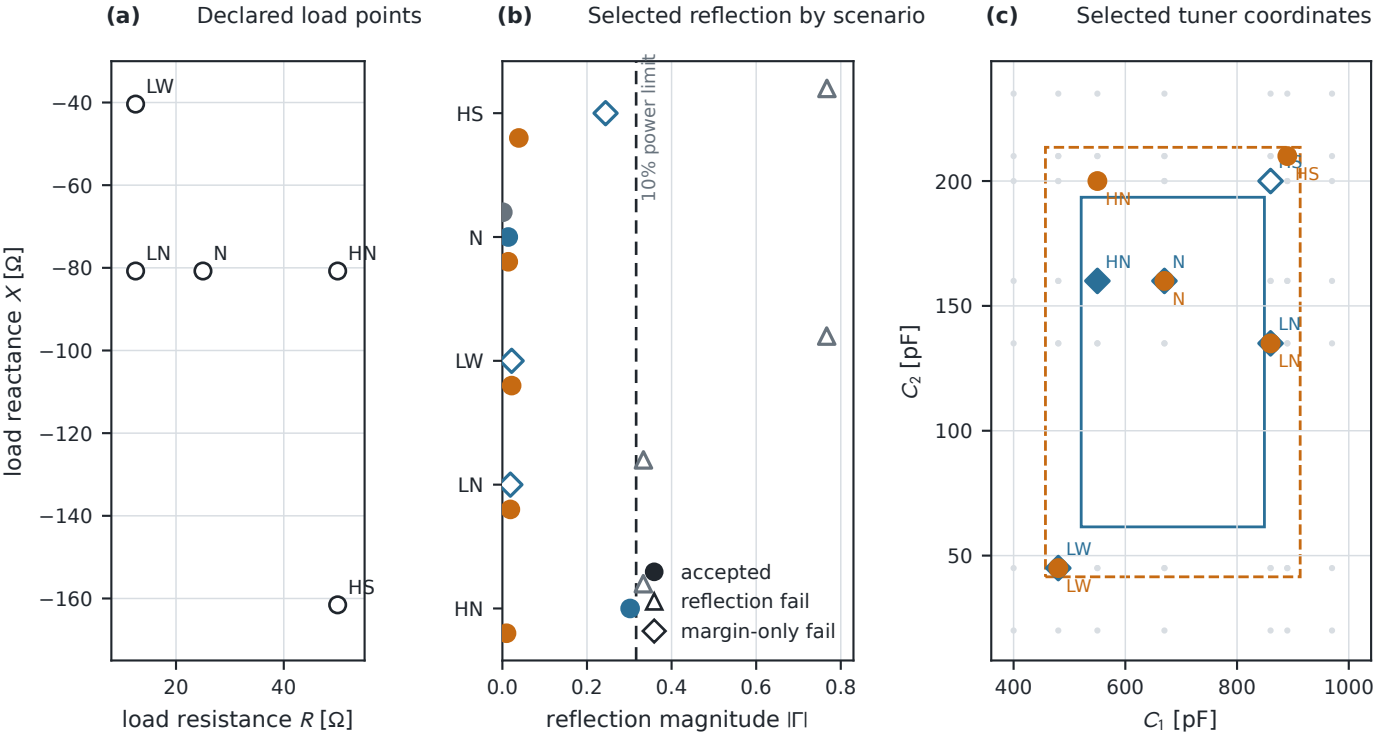
Reference-plane representations used by D1-D3

The same lossy R-L fixture is represented once in D1 and D2; D3 deliberately includes it twice.



Control-authority benchmark (B1-B3)

One five-point synthetic load envelope; B2/B3 select a discrete C1-C2 state independently for each scenario.

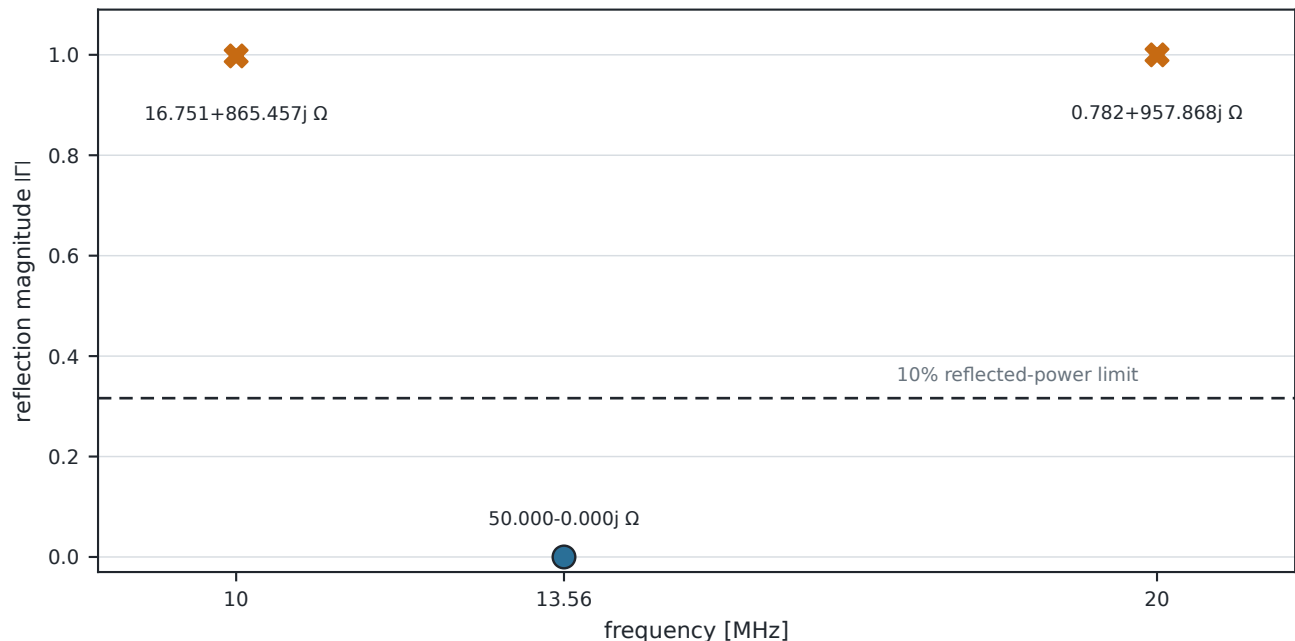


Panel (b): B1 grey, B2 blue, B3 orange; fill/shape gives status.
Panel (c): blue diamonds = B2, orange circles = B3; rectangles give 20% reserve regions.

A5 effective-ICP frequency conformance

Fixed L-match designed at 13.56 MHz; three synthetic terminal-model conditions are evaluated independently.

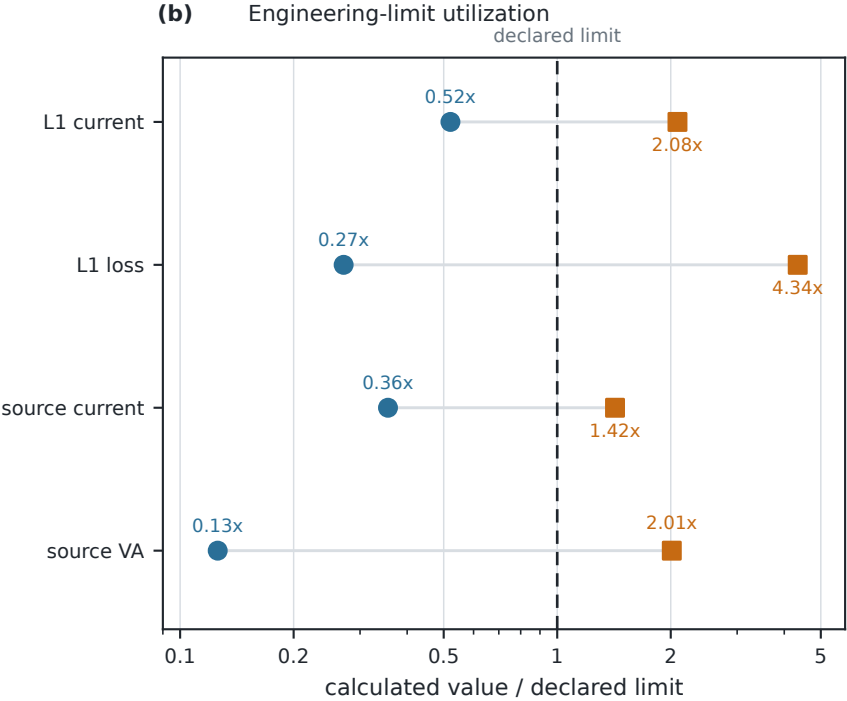
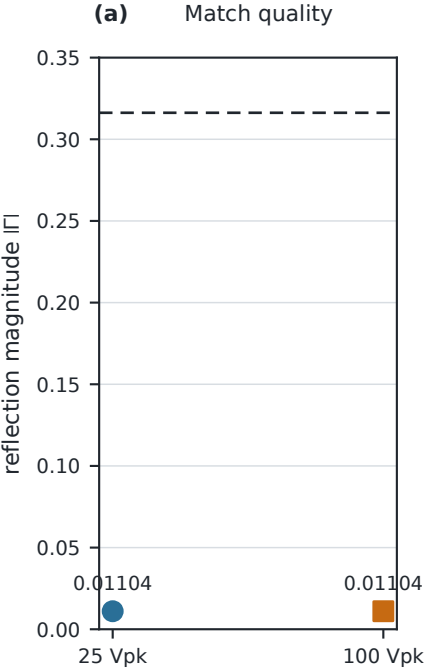
(a) Independent frequency points



Markers: real ngspice results. No line is drawn because the benchmark does not claim a continuous qualified bandwidth.

B5 high-drive component and source stress

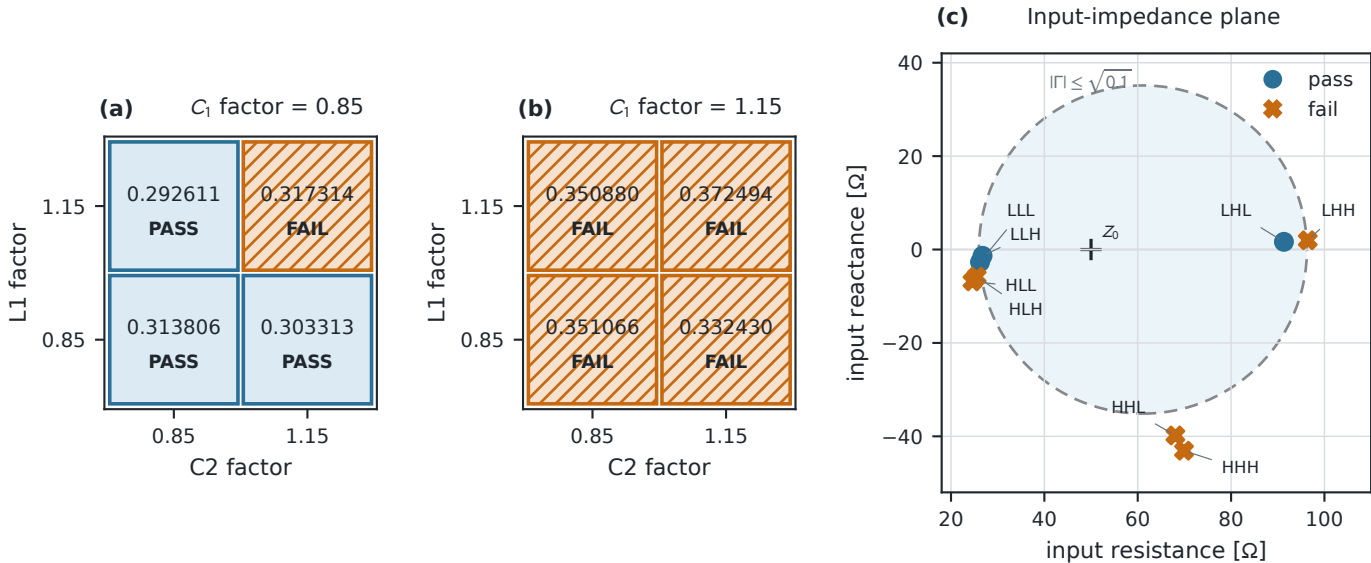
The linear circuit remains matched at both amplitudes; declared L1 and source-terminal limits reject the high-drive scenario.



Blue circle = 25 Vpk; orange square = 100 Vpk. Utilization = calculated value / limit.
Effective ESR/DCR loss is not a temperature or lifetime prediction.

B8 deterministic component-value corners

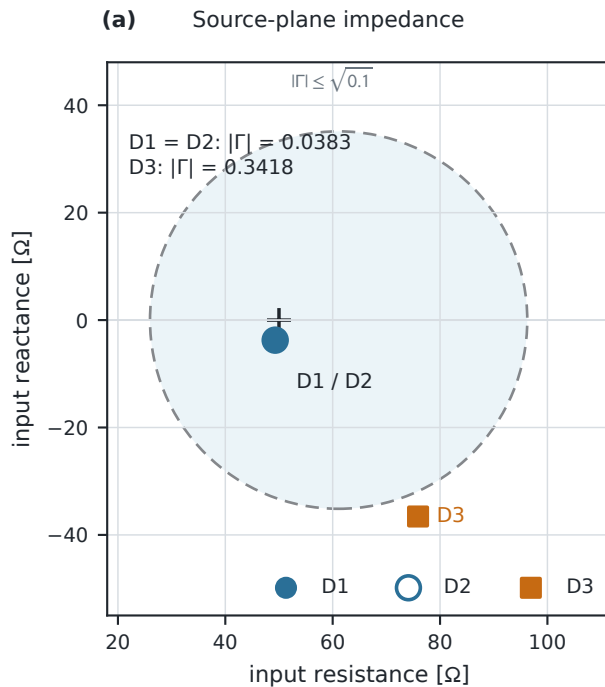
Nominal $C_1/L_1/C_2$ Candidate with the complete 0.85/1.15 factor product; cell values are $|Γ|$.



Three of eight vertices pass. Point labels give $C_1/L_1/C_2$ as L = 0.85 or H = 1.15; the count is corner coverage, not manufacturing yield.

D1-D3 reference-plane benchmark results

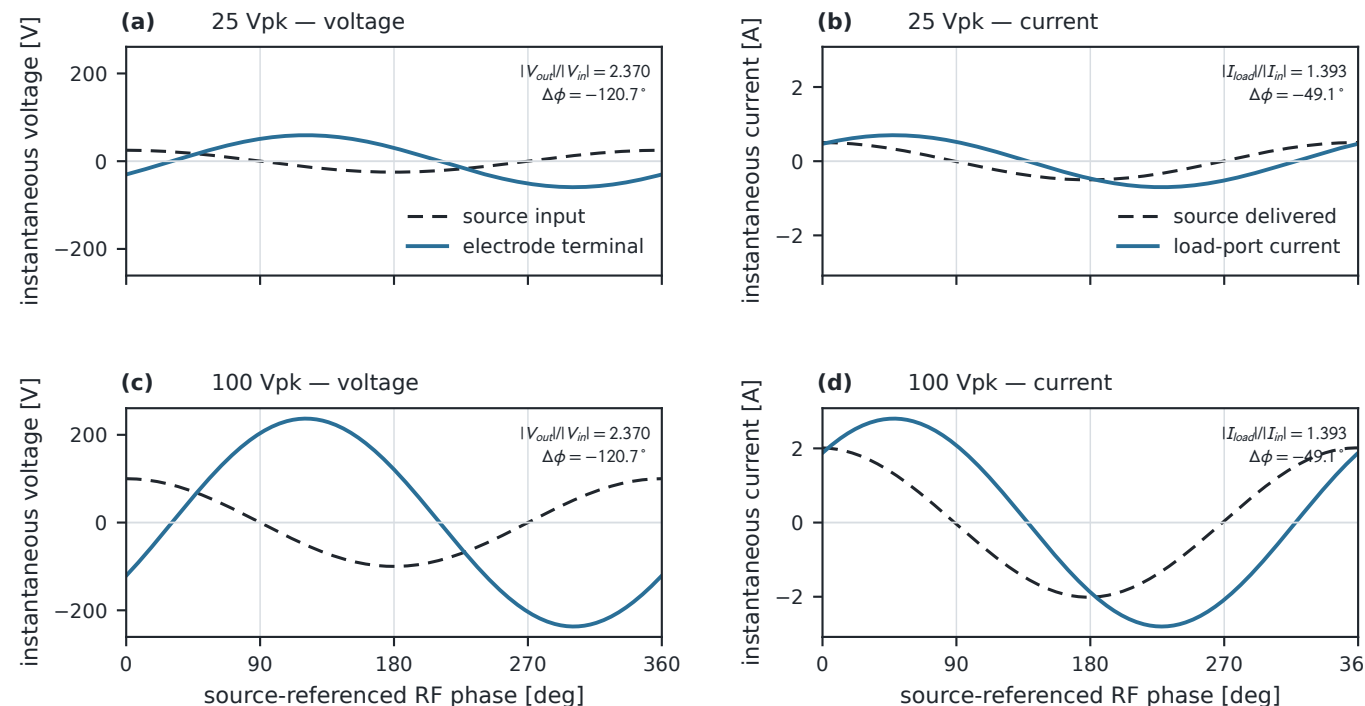
D1/D2 overlap; duplicate fixture D3 fails the 10% reflected-power limit.



Source evaluation plane; the blue circle is the exact $Z_0 = 50\ \Omega$, $|\Gamma| \leq \sqrt{0.1}$ region.

B5 source and electrode-terminal steady-state fundamental

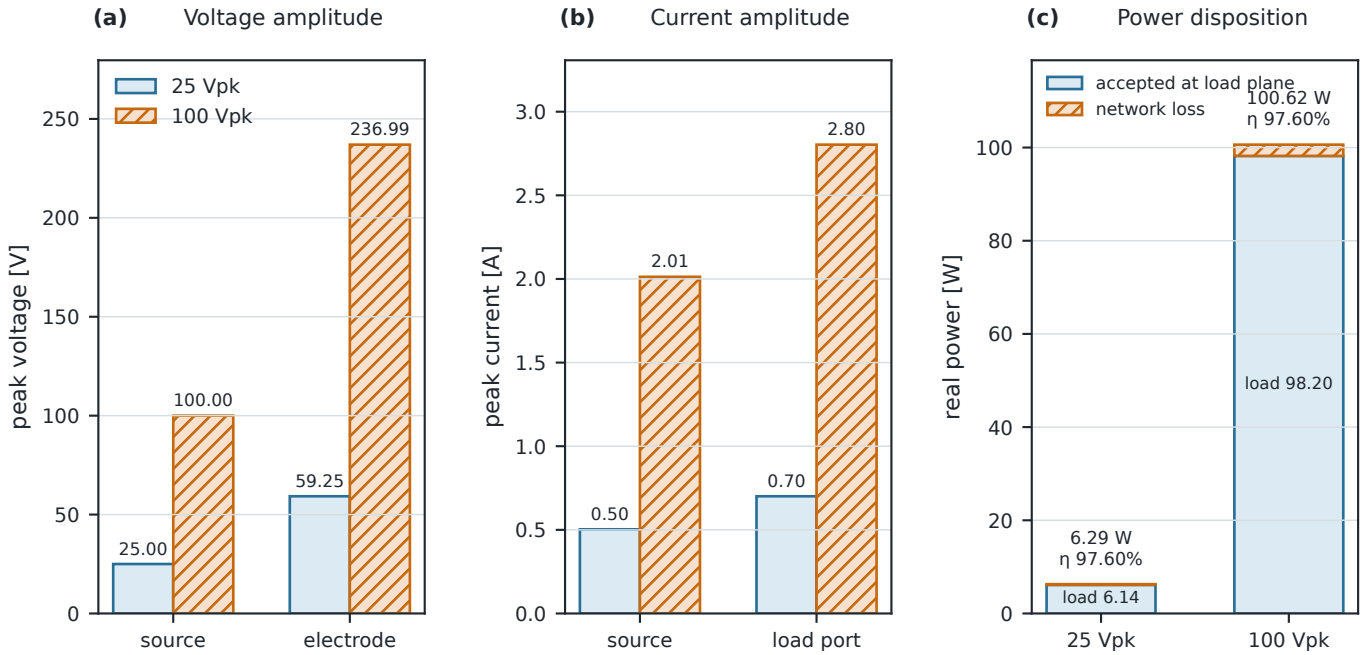
Reconstructed from one 13.56 MHz ngspice AC phasor per drive condition; these curves are not transient or harmonic results.



Input = ideal-source node voltage and delivered current.
Output = declared electrode_terminal voltage and load current; AC magnitudes are peak values.

B5 terminal amplitudes and real-power transfer

Two discrete ngspice AC solutions at one fixed linear network/load; bars do not imply a continuous amplitude sweep.

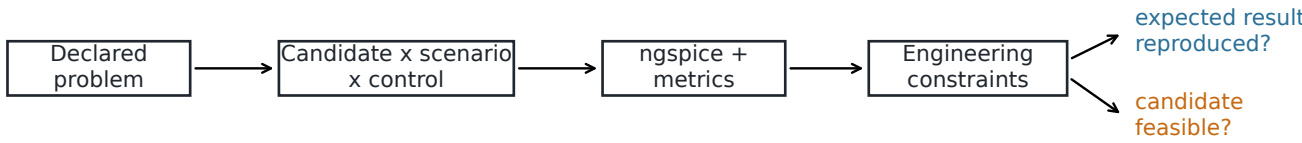


A 4x drive produces 4x terminal amplitudes and 16x power/loss while $|\Gamma|$ and 97.60% efficiency stay constant. The high-drive condition fails declared source/L1 stress limits, not the match criterion.

How to read the core benchmark results

Benchmark pass = declared expectation reproduced; candidate feasible = every required engineering constraint satisfied.

(a) What one benchmark case does



(b) Case outcomes		expectation reproduced	candidate feasible
A1	L-match equation	●	✖
A2	Pi-match equation	●	●
A3	Harmonic branch	●	✖
A4	Effective CCP load	●	✖
A5	Effective ICP load	●	✖
B1	Fixed nominal	●	✖
B2	Limited tuner	●	✖
B3	Full tuner	●	●
B4	Frequency points	●	✖
B5	Drive stress	●	✖
B6	Hardware shortlist	●	●
B7	Role factorial	●	●
B8	Value corners	●	✖
D1	Explicit fixture	●	●
D2	Embedded fixture	●	●
D3	Double-counted fixture	●	✖

Frozen run: all 16 benchmark expectations reproduced.
Engineering outcome: 6 candidate configurations feasible; 10 intentionally or diagnostically infeasible.